

Decent Coaching Classes

X Semi

Mind Your Math - V

ANSWER KEY

Time: 1 Hour

Total Marks: 30

Note: Marks for each question are shown in []. Any other correct method should be given full credit. Answers in bold are the expected fills / final answers.

Q.1 (A) Select the correct alternative

4 Marks

- (1) (C) 3 $3(2) + 2y = 12 \therefore 2y = 6 \therefore y = 3$ [1]
(2) (D) -2 $y = D_y / D = -8 / 4 = -2$ [1]
(3) (A) $x^2 - 4x + 4 = 0$ It is of the form $ax^2 + bx + c = 0$ with $a \neq 0$ [1]
(4) (C) not real If $b^2 - 4ac < 0$, the roots are not real [1]

Q.1 (B) Solve any four

4 Marks

- (1) Value of determinant $= (3 \times 5) - (2 \times 1) = 15 - 2 = \underline{13}$ [1]
(2) $x(x + 3) = 10 \therefore x^2 + 3x = 10 \therefore \underline{x^2 + 3x - 10 = 0}$ [1]
(3) $a = 1, b = -6, c = 5 \therefore b^2 - 4ac = 36 - 20 = \underline{16}$ [1]
(4) $a_1 = 2, b_1 = 1, a_2 = 1, b_2 = -1 \therefore D = a_1b_2 - a_2b_1 = (2)(-1) - (1)(1) = \underline{-3}$ [1]
(5) Put $x = 2: x^2 - 5x + 6 = 4 - 10 + 6 = 0 = \text{RHS} \therefore x = 2$ is a root [1]

Q.2 (A) Activities (any two)

4 Marks

- (1) $4x + y = 14 \dots (I)$ $2x - y = 4 \dots (II)$ [2]
Adding (I) and (II): $6x = \underline{18} \therefore x = \underline{3}$
Putting $x = 3$ in (I): $4 \times \underline{3} + y = 14 \therefore y = 14 - \underline{12} = \underline{2}$
 $\therefore$ Solution is $(x, y) = (\underline{3}, \underline{2})$.
(2) $a_1 = 1, b_1 = 2, c_1 = 8; a_2 = 2, b_2 = -1, c_2 = 1$ [2]
 $D = a_1b_2 - a_2b_1 = (1)(-1) - (2)(2) = \underline{-5}$
 $D_x = c_1b_2 - c_2b_1 = (8)(-1) - (1)(2) = \underline{-10}$
 $D_y = a_1c_2 - a_2c_1 = (1)(1) - (2)(8) = \underline{-15}$
 $x = D_x / D = \underline{-10} / -5 = \underline{2}$ $y = D_y / D = -15 / \underline{-5} = \underline{3}$
(3) $x^2 + 7x + 12 = 0$ [2]
 $x^2 + 3x + \underline{4}x + 12 = 0$
 $x(x + 3) + \underline{4}(x + 3) = 0$
 $(x + 3)(x + \underline{4}) = 0$
 $\therefore x + 3 = 0$ or $x + 4 = 0 \therefore x = \underline{-3}$ or $x = \underline{-4}$

Q.2 (B) Solve any four

8 Marks

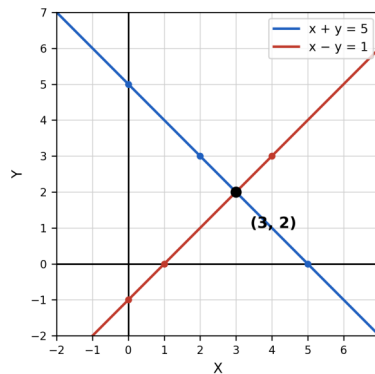
- (1) $3x + 2y = 14 \dots (I)$ $x - 2y = 2 \dots (II)$ [2]
Adding (I) and (II): $4x = 16 \therefore x = \underline{4}$
Putting $x = 4$ in (II): $4 - 2y = 2 \therefore -2y = -2 \therefore y = \underline{1} \therefore (x, y) = (\underline{4}, \underline{1})$
(2) $a_1 = 2, b_1 = 1, c_1 = 5; a_2 = 1, b_2 = -1, c_2 = 1$ [2]
 $D = a_1b_2 - a_2b_1 = (2)(-1) - (1)(1) = \underline{-3}$
 $D_x = c_1b_2 - c_2b_1 = (5)(-1) - (1)(1) = \underline{-6}$
 $D_y = a_1c_2 - a_2c_1 = (2)(1) - (1)(5) = \underline{-3}$
(3) $x^2 - 9x + 20 = 0 \therefore x^2 - 4x - 5x + 20 = 0$ [2]
 $\therefore x(x - 4) - 5(x - 4) = 0 \therefore (x - 4)(x - 5) = 0$
 $\therefore x = \underline{4}$ or $x = \underline{5}$
(4) Comparing with $ax^2 + bx + c = 0: a = 1, b = 4, c = 5$ [2]
 $b^2 - 4ac = 16 - 20 = -4 < 0 \therefore$ the roots are not real.

- (5) Roots are 2 and 5. Sum of roots = $2 + 5 = 7$ and product of roots = $2 \times 5 = 10$ [2]
Equation: $x^2 - (\text{sum of roots})x + \text{product of roots} = 0 \therefore \underline{x^2 - 7x + 10 = 0}$

Q.3 Solve any two

6 Marks

- (1) Table of values (1 mark for each equation, 1 mark for graph and solution): [3]
 $x + y = 5$: $x = 0, y = 5 \rightarrow (0, 5)$; $x = 5, y = 0 \rightarrow (5, 0)$; $x = 2, y = 3 \rightarrow (2, 3)$
 $x - y = 1$: $x = 0, y = -1 \rightarrow (0, -1)$; $x = 1, y = 0 \rightarrow (1, 0)$; $x = 4, y = 3 \rightarrow (4, 3)$



The two lines intersect at the point (3, 2). $\therefore$ Solution is $(x, y) = (\underline{3}, \underline{2})$.

- (2) $2x + 3y = 9, x - y = 2$: $a_1 = 2, b_1 = 3, c_1 = 9$; $a_2 = 1, b_2 = -1, c_2 = 2$ [3]
 $D = a_1b_2 - a_2b_1 = (2)(-1) - (1)(3) = -2 - 3 = -5$
 $D_x = c_1b_2 - c_2b_1 = (9)(-1) - (2)(3) = -9 - 6 = -15$
 $D_y = a_1c_2 - a_2c_1 = (2)(2) - (1)(9) = 4 - 9 = -5$
 $x = D_x / D = -15 / -5 = \underline{3}$ $y = D_y / D = -5 / -5 = \underline{1}$ $\therefore (x, y) = (\underline{3}, \underline{1})$
- (3) $2x^2 + 5x - 3 = 0$. Comparing with $ax^2 + bx + c = 0$: $a = 2, b = 5, c = -3$ [3]
 $b^2 - 4ac = 25 - 4(2)(-3) = 25 + 24 = 49$
 $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-5 \pm \sqrt{49}}{4} = \frac{-5 \pm 7}{4}$
 $\therefore x = \frac{-5 + 7}{4} = \underline{1/2}$ or $x = \frac{-5 - 7}{4} = \underline{-3}$

Q.4 Solve any one

4 Marks

- (a) Let the father's age be x years and the son's age be y years. [4]
 $x + y = 45 \dots (I)$ $2x + y = 80 \dots (II)$ (2 marks for framing both equations)
Subtracting (I) from (II): $x = 35$
Putting $x = 35$ in (I): $35 + y = 45 \therefore y = 10$
 $\therefore$ Present age of the father = 35 years and of the son = 10 years.
- (b) Let the breadth of the rectangle be x m. Then the length = $(x + 3)$ m. [4]
Area = length $\times$ breadth $\therefore x(x + 3) = 40 \therefore x^2 + 3x - 40 = 0$
 $x^2 + 8x - 5x - 40 = 0 \therefore x(x + 8) - 5(x + 8) = 0 \therefore (x + 8)(x - 5) = 0$
 $\therefore x = -8$ or $x = 5$. Breadth cannot be negative, so $x \neq -8 \therefore x = 5$
 $\therefore$ Breadth = 5 m and length = $5 + 3 = \underline{8 \text{ m}}$.